

P. pavoninus-Inspired Smart Slips Marine Antifouling Coating Based on Coumarin: Antifouling Durability and Adaptive Adjustability of Lubrication

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Slippery liquid-infused porous surface (SLIPS) is received widespread attention in the antifouling field because of its dynamic slippery surface, but the limited effect of physical “anti-adherence” and the problem of lubricant loss restrict its application for practical marine antifouling. Inspired by the self-defensive function of *P. pavoninus*'s poisonous mucus, a SLIPS based on the reversible chemical bonds and the antibacterial effect of coumarin (CmSLIPS) is prepared, which addresses these disadvantages by responsively adaptive lubricity and durability. The CmSLIPS enables the “locking” and “unlocking” mode between lubricant and matrix via the reversible photodimerization of coumarin. It regulates the surface lubricity to adapt to the antifouling demand. The reversible chemical bonds of lubricant and matrix reduce lubricant loss and enhance the lubricant stability and antifouling durability. In addition to the physical anti-adhesive barrier of SLIPS, the excellent antimicrobial property of coumarin groups further enhances the “biocidal” antifouling effect to improve long-term antifouling durability. In summary, the CmSLIPS has surface stability, durability, responsively adaptive lubricity, efficient self-cleaning, anti-protein, anti-bacterial, anti-algae, self-healing (92.85%) properties and 150-day marine field antifouling performance during boom seasons, which maintains the satisfactory antifouling performance in the harsh marine environment and demonstrates the promising application in neritic sea equipment.

1. Introduction

Biofouling always has seriously negative effects on many aspects of human being lives,^[1] especially doing great harm to maritime affairs.^[2,3] The antifouling coating is one of the most effective and feasible methods to avoid marine biofouling.^[4] As tributyltin (TBT) was forbidden globally in marine coatings due to its teratogenicity,^[5] eco-friendly and effective marine antifouling coatings are still extensively studied.^[6]

Many antifouling coatings have been developed for the battle against marine biofouling. The antifouling strategies are mainly divided into two types, the antifouling (AF) “offensive” strategy and the fouling-release (FR) “defensive” strategy.^[1] AF strategy includes self-polishing,^[7] eco-friendly bactericidal,^[8,9] antifoulant release,^[9] dynamic biodegradable,^[7,10,11] and mimic peptide coating.^[12,13] FR strategy includes low-surface-energy (SE),^[14–16] micro-structured,^[17,18] hydrophilic,^[6,19–21] amphiphilic,^[22–24] SLIPS,^[25–28] and fluorescent coating.^[29]

The SLIPS materials have attracted continuous attention because of the low SE, the low Young's modulus and the dynamic slippery lubricant surface.^[30–34] Compared with traditional low SE coatings, the dynamic lubricating layer significantly prevents biofouling adhesion.^[35] However, the limited physical “anti-adherence” effect has insufficient static antifouling effect in harsh marine environments.^[36] Most of lubricants for SLIPS are silicone oil, fluoro-ether or liquid paraffin, which do not exhibit biocide effect.^[32,37] The loss of lubricant also restricts the lifespan of SLIPS in real sea.^[38] In addition, the traditional marine coatings have only one single antifouling mode, which cannot meet the needs of different times and spaces.^[39] Therefore, the “smart” coatings that can respond to environmental stimuli and adjust the antifouling performance have high research value.^[40]

Herein, inspired by the defensive function of *P. pavoninus*'s poisonous mucus,^[41] the lubricant and matrix of the SLIPS were modified with coumarin to prepare a “smart” SLIPS antifouling coating (CmSLIPS), which has adaptive adjustability of lubrication^[42–44] and antifouling durability.^[45–47] Lubricant loss

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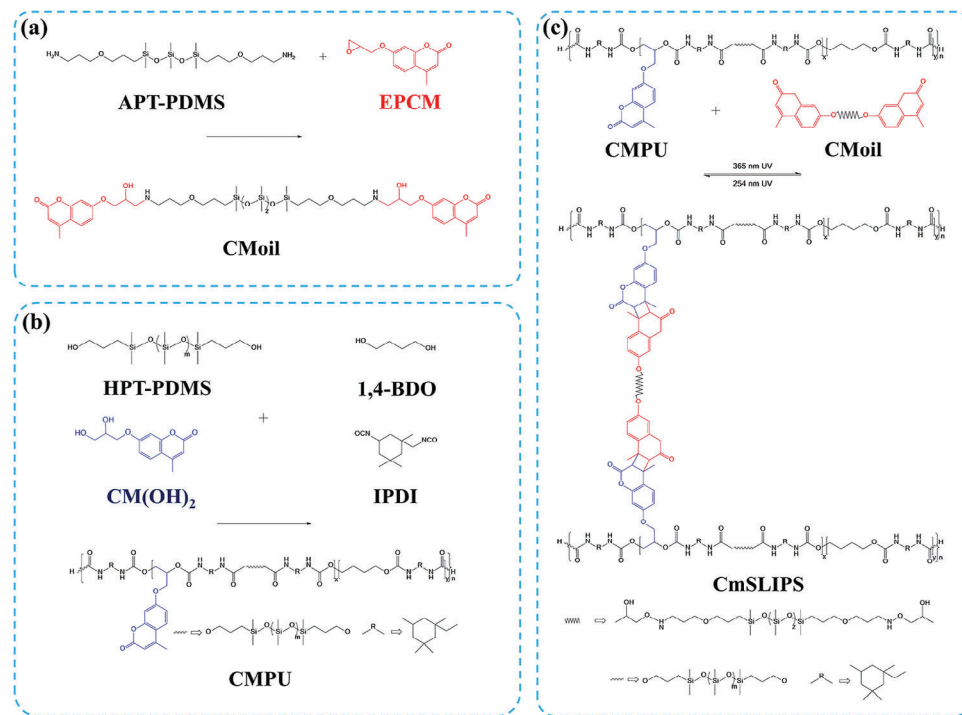
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Scheme 1. a) Synthesis of CMoil, b) CMPU, and c) CmSLIPS.

and insufficient physical antifouling effect restrict the development of SLIPS materials in real marine environments. The *P. pavoninus* can synthesize toxic mucus and regulate its secretion to avoid the attack of sharks. The active ingredient—pardaxin has been used as a shark repellent.^[48–50] Similarly, in addition to the physical anti-adhesive lubrication barrier, the excellent antimicrobial property of coumarin groups further enhances the antifouling effect of CmSLIPS. “FR” and “AF” antifouling strategies are successfully integrated into the CmSLIPS to improve the antifouling efficiency and durability. The CmSLIPS modifies the surface lubricity via modulating lubricant behavior. The reversible chemical bonds of coumarin in matrix and lubricant reduce the lubricant loss and adjust the lubricant mobility. The coumarin-modified lubricant (CMoil) is previously locked in the coumarin-modified matrix (CMPU) via photo-dimerization under 365 nm UV. The dimerization connection effectively reduces the lubricant loss. With the simulation of 254 nm UV, the dimerization bonds between coumarins are disconnected, which raise the mobility of lubricant. The “unlocking” effect between CMoil and CMPU enhances surface lubricity to strengthen antifouling property. The CmSLIPS adjusts the reversible “lock” between the lubricant and matrix via the reversible photodimerization of coumarin, thus regulating surface lubricity to adapt to different antifouling needs. In addition, the coumarin groups introduce the “biocide” antifouling effect to the CmSLIPS. De-dimerization does not occur in the dark. The lubricant mobility is weak, but sufficient to cope with the weaker biofouling. The coumarin also de-dimerized under sunlight. As the cyclobutane structure formed by the coumarin dimer is thermodynamically unstable and is susceptible to jump to low energy level of monomers via 254 nm UV in sunlight, the efficiency of photo-depolymerization

is higher than that of photo-dimerization under sunlight. Surface lubricant mobility increases, and the enhanced surface lubricity is able to cope with the vigorous growth of marine biofouling during the daytime. The CmSLIPS has antibacterial property, adaptive antifouling performance and 150-day antifouling performance in actual marine fields during boom seasons, which proves its efficient antifouling effect on neritic sea equipment in harsh marine environments.

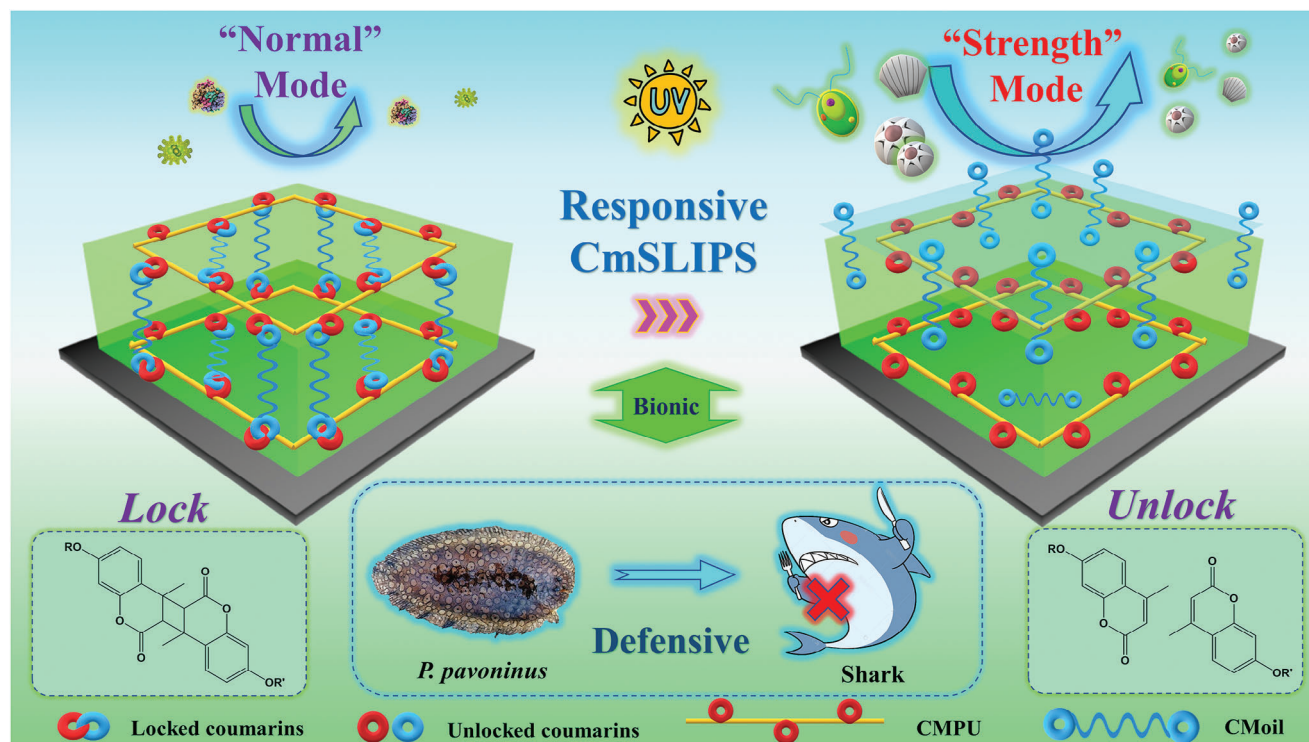
2. Results and Discussion

2.1. Synthesis Process

The synthesis process of dihydroxy-coumarin (CM(OH)₂), epoxy-coumarin (EPCM), CMPU and CMoil was shown in Schemes S1 and S2 (Supporting Information) and **Scheme 1**. CM(OH)₂ was synthesized into CMPU as side chain groups. The epoxy group of EPCM reacted with the amino group of aminopropyl terminated poly(dimethylsiloxane) to synthesize the coumarin-capped silicone oil—CMoil. The successful synthesis of CM(OH)₂, EPCM, CMPU, and CmSLIPS could be certified by the ¹H NMR and FTIR in Figure S1–S7 (Supporting Information). The CmSLIPS was labeled as CmSLIPS-0/10/20/30 according to the amount of CMoil (**Scheme 2**).

2.2. Surface Property and Morphology

The surface properties of coatings, such as lubricity, water contact angle (WCA), and surface energy (SE), have a significant impact



Scheme 2. Design and responsive mechanism of the CmSLIPS antifouling coating.

on the antifouling performance.^[51] Surface lubricity was characterized by the sliding speed of water droplets and WCA on the coating. Before 254 nm UV irradiation, the WCA was about 107–109°, which proved the hydrophobicity of CmSLIPS and the ability to apply in marine environments. According to the research of Baier,^[52] the surfaces with SE of 20–30 mJ m⁻² are difficult to adhere via biofouling. As Figure 1b shows, The SE of CmSLIPS was in the interval of 20–30 mJ m⁻².^[53] Surface lubricity was characterized by the sliding speed of water droplets on the coating. The sliding velocity increased from about 110 to 170 mm s⁻¹ with the increase of CMoil. The water droplet took only 0.42 s to slide off from the CmSLIPS-30 (76 mm length). The CmSLIPS-0 needed 8.64 s (Figure 1a). It indicated that the CmSLIPS-10/20/30 had excellent surface lubricity.

The morphology of the CmSLIPS was characterized by a scanning electron microscope (SEM) and atomic force microscope (AFM) (Figure 2b,c). After the CmSLIPS was irradiated by 365 nm UV, photodimerization occurred between the coumarin groups in the CMPU and CMoil. The closer the coating was to the UV source, the faster the crosslinking speed was, and the strength of the coating after crosslinking by photodimerization increased. As a result, stress unevenness occur in the normal direction of the coating, and the uneven wrinkle microstructure was formed on the surface.^[54] When the CmSLIPS was treated with 254 nm UV, the dimerization between coumarin groups was untied, and the stress unevenness of the coating was improved. After the CMoil was “unlocked” from the CMPU, its mobility was enhanced to migrate to the surface. Therefore, the surface roughness was reduced, and the uneven microstructure disappeared. The reduction of surface roughness and the formation of the lu-

bricating layer effectively enhanced the lubricating property of the coating. The change of the surface topography after the response proved the ability to responsively regulate the surface and morphology lubricity.

2.3. Adaptive Adjustability of Lubrication

The CmSLIPS responded to light stimuli and adaptively adjusting surface lubricity, and adaptively adjusted the surface lubricity in demand. As shown in the FTIR (Figure 2a), the disappearance of the 1600 cm⁻¹ peak of C=C proved that the coumarin groups in the CMPU and CMoil were dimerized. The reappearance of the 1600 cm⁻¹ peak after 254 nm UV confirmed the de-dimerization of coumarin groups in CMPU and CMoil.

After 254 nm UV irradiation, coumarin groups of CMoil and CMPU de-dimerized on the surface. The CMoil increased mobility and migrated on the surface. The lubricity increased. The surface roughness and WCA were reduced. According to Wenzel–Cassie theory,^[55,56] the decrease in surface roughness leads to the decrease in hydrophobicity. The decrease of the WCA and increase of drops sliding speed after 254 nm UV (Figure 1a,c) proved the “unlock” of CMoil. As shown in Figure 1d, the droplet slip rate after response increased and the surface lubricity was enhanced with the CMoil content.

The responsive behavior of CMoil was also characterized by the surface microscopic morphology and energy dispersive spectrometer (EDS) (Figure 2b,d). After 254 nm UV irradiation, the irregular wrinkle pattern caused by the photo-dimerization crosslinking disappeared. The stress unevenness decreased

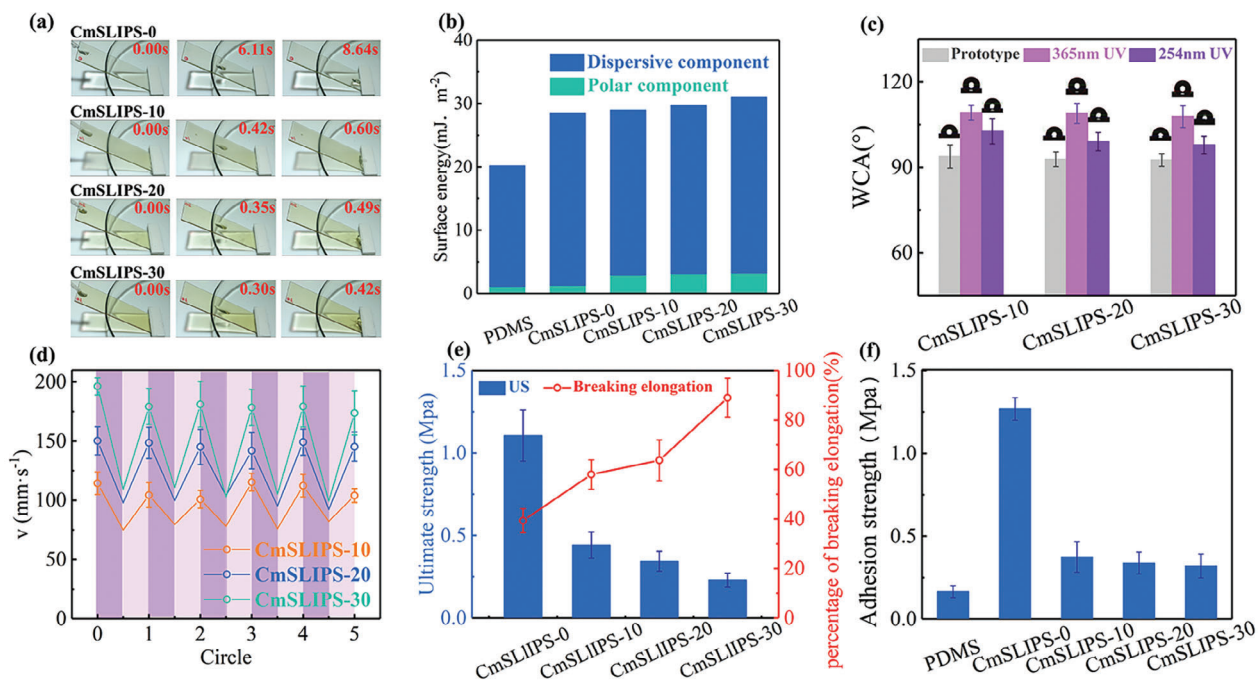


Figure 1. a) Water drops sliding performance; b) surface energy; c) WCA; d) lubricant self-replenishment cycle test; e) ultimate strength and breaking elongation percentage; f) adhesion strength.

after de-dimerization. The unlocked CMoil migrated to the surface and formed a liquid lubricating layer. As the SEM and AFM images show, the irregular surface wrinkle structure obviously disappeared due to the unlocked CMoil. The EDS results showed that the content of N and O elements increased and the content of Si element decreased after 254 nm UV irradiation. This was because the CMoil had larger atomic percentage of N and O and smaller atomic percentage of Si than that of CMPU. As the CMoil was unlocked with CMPU, more CMoil molecular migrated to

the surface and formed a lubricating layer. The increased proportion of CMoil on the surface caused changes in the content of surface elements.

2.4. Surface Stability and Durability

The conventional SLIPS materials have the problem of lubricant loss, which will lead to the decrease in surface lubricity and

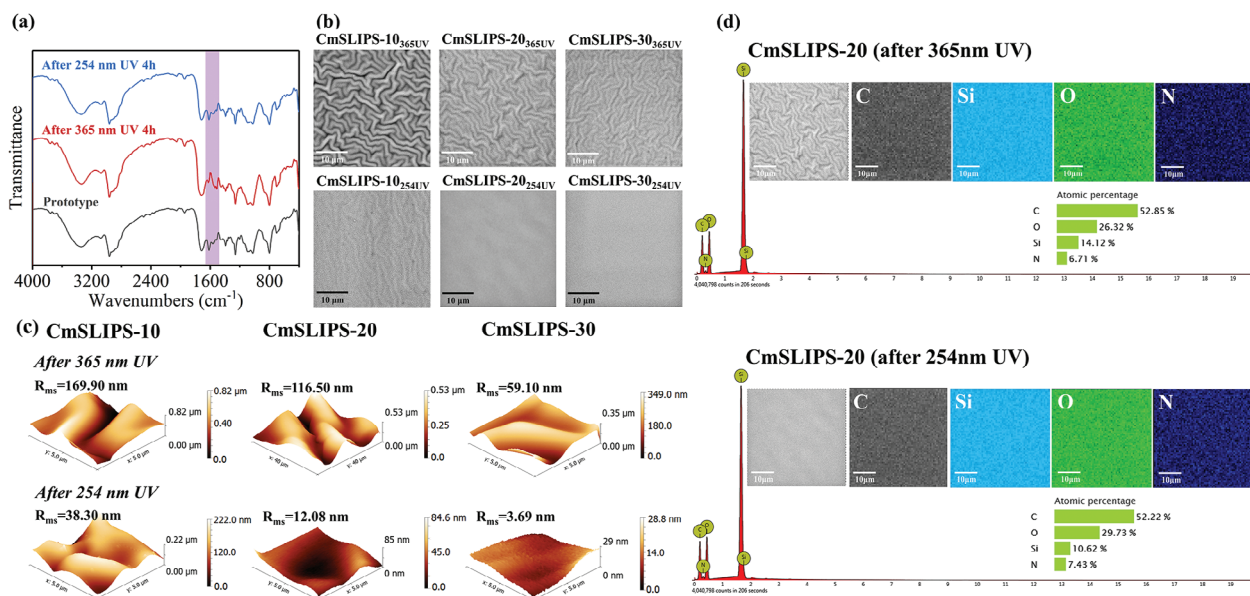


Figure 2. a) FITR, b) SEM, c) AFM, and d) EDS of the CmSLIPS after UV response.

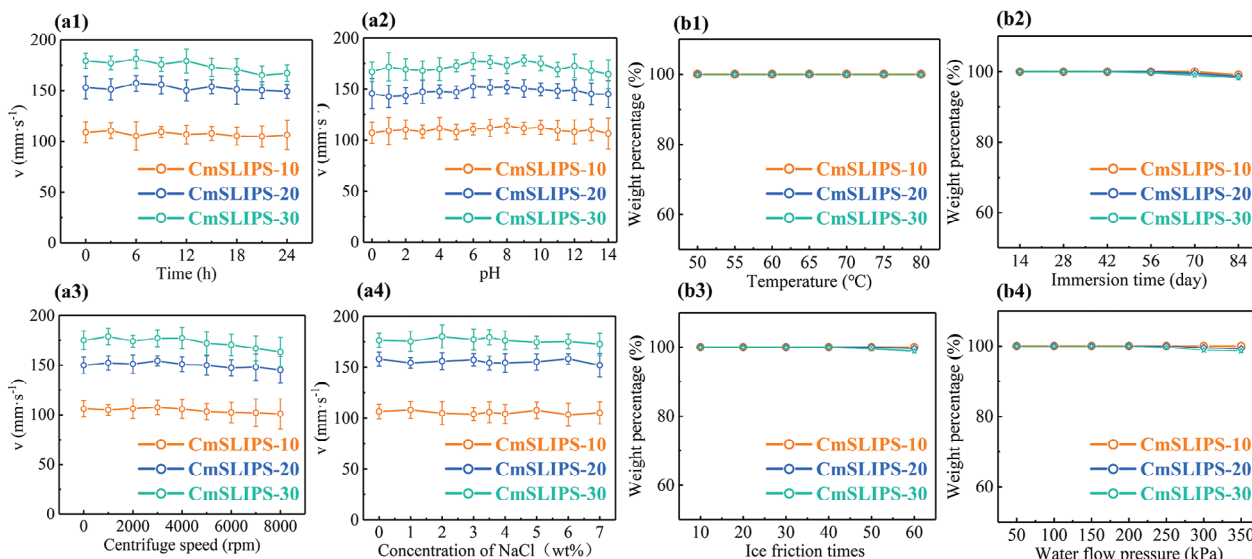


Figure 3. a1) Lubricity stability for immersion, a2) pH, a3) centrifugal rotation and a4) salinity; b1) anti-loss durability of lubricant for heating, b2) long-period immersion, b3) ice friction, and b4) water impact.

antifouling performance. Similar compatibility between silicone CPU and silicone CMoil enhanced the stability of the lubricant. Due to the chemical stability of silicone, the surface lubricity of CmSLIPS was not affected in artificial seawater (ASW) with different pH (0–14) and salinity (0–7.0 wt%) (Figure 3a). Due to the compatibility of CPU and CMoil and the entanglement of polymer chains, the surface lubricity of the CmSLIPS did not decrease significantly under the action of 400 rpm rotating water flow or centrifugation (1000–8000 rpm). It demonstrated that CmSLIPS had excellent surface lubricity stability in “unlocked” mode.

However, in face of harsh external environment, it is still necessary to enhance the anti-loss performance of lubricant in SLIPS. The CmSLIPS effectively anchored the lubricant in the matrix to prevent loss via the reversible chemical bonds among the coumarin groups. As the CMoil was connected with the CPU matrix via reversible covalent bonds of coumarin, external force, heat or long-term immersion did not make the lubricating fluid easily lost. As shown in Figure 3b, after heating (50–80 °C), water impact with a certain pressure (0.1–0.35 Mpa), ice cube (5 g) friction and long-term immersion (84 days), the quality of CmSLIPS did not decrease significantly. It showed that the “locking” effect by the photo-dimerization of coumarin prevented the lubricant loss and demonstrated durability under severe conditions (such as high temperature, ice floes, sea waves and long-term immersion). The CmSLIPS also has cyclic self-replenishment property (Figure 1d). After the CMoil on the surface was wiped and removed by oil-absorbing paper, the CMoil in the CPU matrix unlocked and migrated to the surface to restore the lubricity after the heat and 254 nm UV irradiation.

2.5. Mechanical and Self-Healing Performance

It has proved that the surface of lower modulus was easy to prevent biofouling adhering. The CmSLIPS had a low Young's

modulus below 0.90 Mpa, which was feasible to reduce biofouling adhesion^[53] (Figure 4a). Breaking elongation (57.97–88.97%) showed the flexibility of CmSLIPS (Figure 1e). The adhesion of CmSLIPS to the epoxy substrate was up to 0.35 Mpa. The adhesion of matrix CPU (CmSLIPS-0) was about 1.26 Mpa, which was robust enough to be applied as marine coatings in the neritic sea^[57] (Figure 1f).

Traditional PDMS was difficult to repair after external mechanical damage.^[58] The reversible covalent bonds effectively improve the self-healing property. The self-healing mechanism is shown in Figure 4a. It is achieved via the dedimerization–dimerization process of coumarin groups and hydrogen bonds on the fracture surface. The reconnected photodimerization bonds effectively enhance the self-healing performance. The “broken hearts” with different colors (Figure 4b) were repaired under UV irradiation. As shown in Figure 4d), the cutting scratches disappeared gradually. The self-healing efficiency reached up to 92.85% (Figure 4a). The CMoil promoted the movement of polymer segments like plasticizer, which increased the reconnection efficiency of photodimer bonds. The self-healing efficiency of CmSLIPS improved with the content of CMoil.

2.6. Antifouling Performance

2.6.1. Self-Cleaning Performance

The self-cleaning performance reflected the ability to resist inorganic marine pollutants such as sand and mud. It was tested by the solid and liquid model contaminants. As shown in Figure 5c and Figure S9 (Supporting Information), liquid pollutants slipped rapidly without residue because of the lubricant repellency. The solid contaminants were easily carried away by sliding water droplets as the lubricating layer barrier reduced the contact of solids with the coating. In contrast, the self-cleaning effect of CmSLIPS-0 was relatively unsatisfactory.

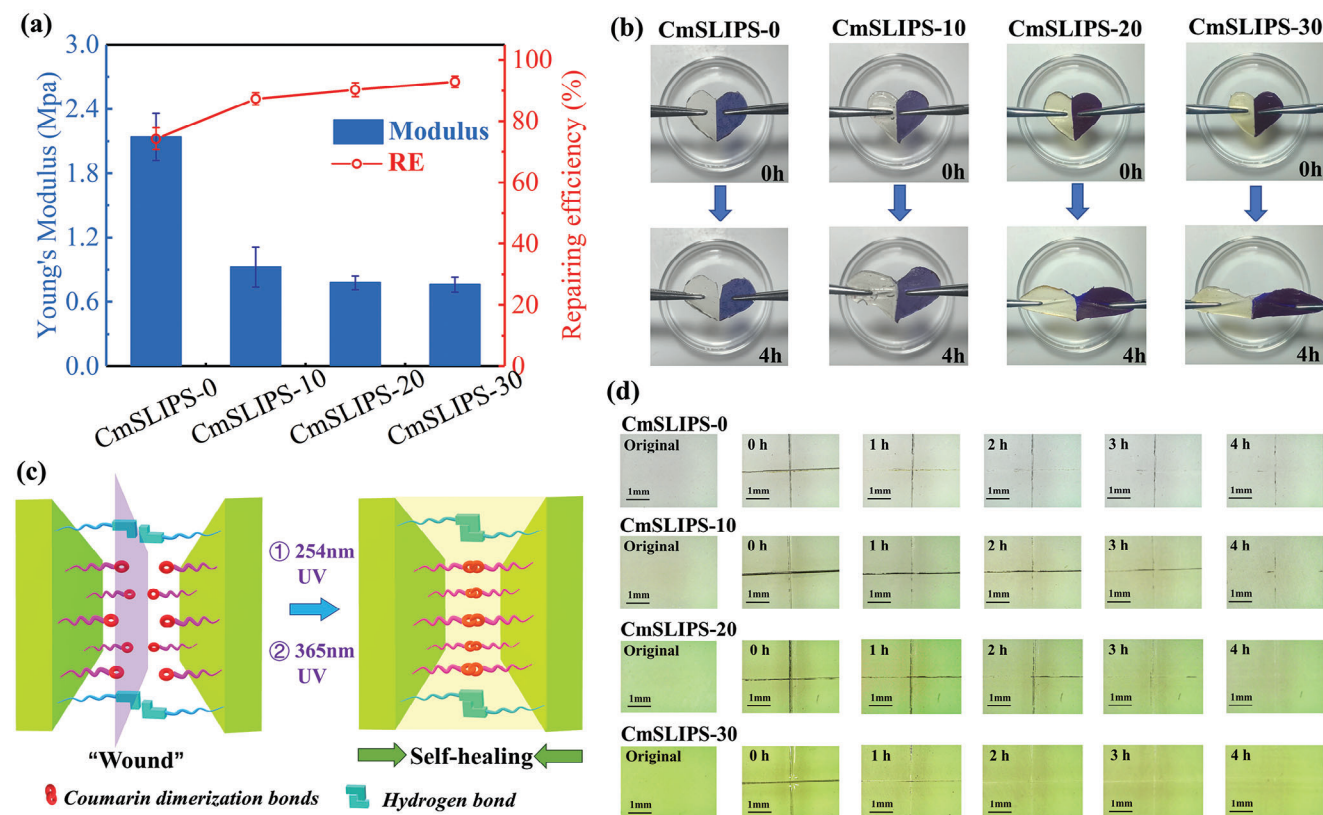


Figure 4. a) Young's modulus and self-healing efficiency of the CmSLIPS; b) digital photos of healed "heart"; c) UV self-healing mechanism; d) digital photos of the UV healing process.

2.6.2. Anti-Protein Performance

Protein is the main reason in the first stage of marine biofouling, which provides a nutrient layer for microorganisms. The lubricating layer reduced the protein's contact and adsorption on the surface. Green fluorescence appeared on the PDMS and CmSLIPS-0, indicating the adhesion of BSA (Figure 5d). CmSLIPS-10/20/30 had almost no green fluorescence, proving the excellent anti-protein performance of CmSLIPS.

2.6.3. Anti-Bacterial and Anti-Algal Performance

Bacteria took the leading part in the second stage of the marine biofouling process. *E. coli* (gram-negative bacteria) and *S. aureus* (gram-positive bacteria) were used as model bacteria to characterize the anti-bacterial performance. The silicone lubricating layer on the CmSLIPS built a dynamic barrier, reducing the contact of bacteria with the matrix. Coumarin compounds competitively inhibit the ATPase's activity of DNA gyrase, making it fail to form DNA double helix structure normally, and inhibit the bacterial quorum sensing (QS) system to avoid the bacterial film.^[45–47] The coumarin groups in CMPU and CMoil killed and evaded bacteria near the surface. Traditional PDMS showed poor anti-bacterial performance. Due to the coumarin antibacterial group contained in CMPU, the antibacterial performance of CmSLIPS-0 was better than PDMS but still unsatisfactory.

CmSLIPS-10/20/30 showed the enhanced antifouling effect with the increase of CMoil addition (Figure 5c).

Algae is also one of the main reasons in the second stage of the marine biofouling process. Anti-algal performance was tested by diatom and chlorella respectively for 14 days (Figure 5a). The anti-algal mechanism of CmSLIPS was similar to the antibacterial mechanism, which could be attributed to the physical barrier of lubricant and the biocide effect of coumarin. Almost no diatoms or chlorella were attached and grown on CmSLIPS-20/30.

2.6.4. Adaptive Antifouling Effect by Lubrication Adjustability

As a "smart" antifouling coating, the CmSLIPS responsively controlled the surface lubrication to adjust the antifouling effect in demand, thereby enhancing antifouling accuracy and prolonging service life. The response mechanism is shown in Figure 6c. Marine fouling organisms thrive during the day and require stronger antifouling effect to cope. The CMoil on the surface of CmSLIPS was unlocked via light stimulation, which continuously enhanced the surface lubricity to achieve a better antifouling effect. In the dark, CMoil was no longer unlocked and migrated, but the antibacterial effect of the coumarin groups was also sufficient to meet the lower antifouling requirement. By adaptively adjusting the activity of the lubricant according to different environments, the loss and waste of the lubricant were reduced. Combining the anti-adhesion effect of slippery surface with the

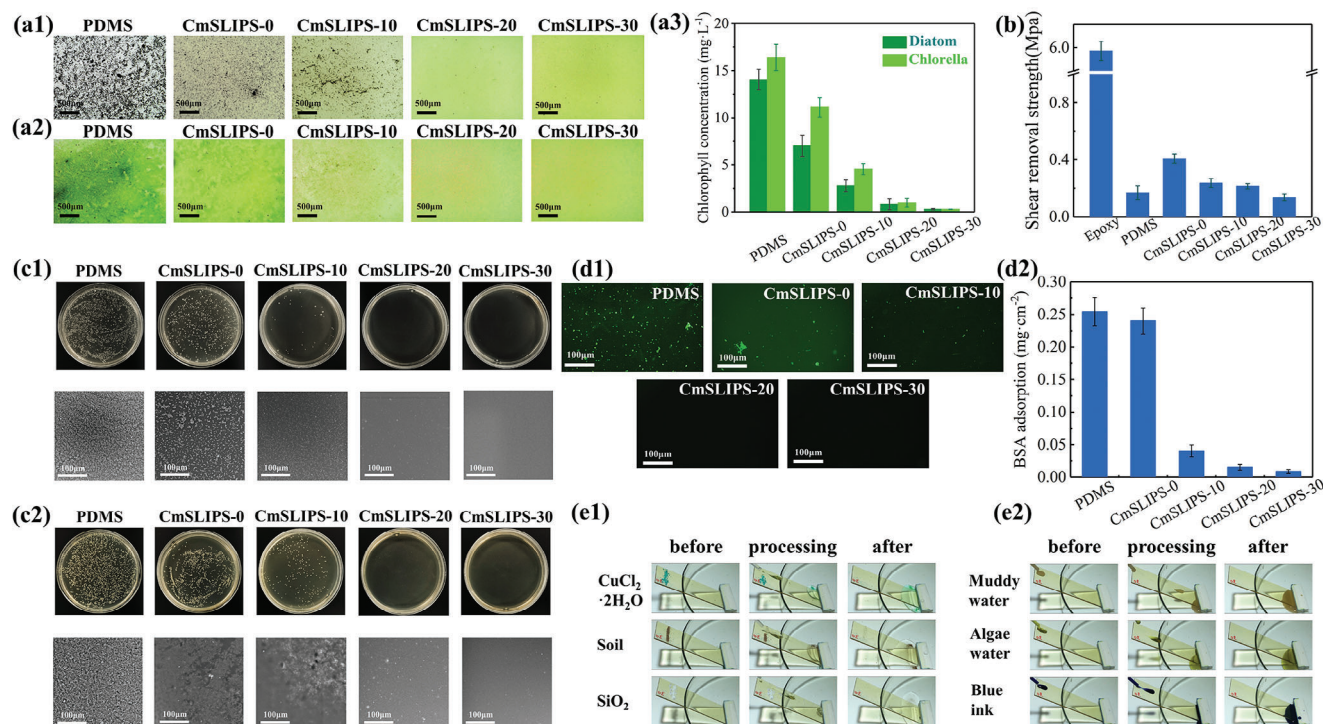


Figure 5. a1) Anti-algae adhesion performance of coatings for 14 days (cycling of 12 h sunlight and 12 h darkness): diatom, a2) chlorococci; a3) quantitative anti-algae characterization based on chlorophyll. b) Pseudobarnacle removal performance test. c1) Anti-*E. coli* and c2) anti-*S. aureus* performance for 24 h (12 h sunlight and 12 h darkness) via LB plate method and SEM observation. d1) Anti-protein (BSA) adhesion performance: fluorescence microscope photos and d2) absorbance-concentration quantitative test. e1) Self-cleaning performance of CmSLIPS-30 against solid model contaminants and e2) liquid model contaminants.

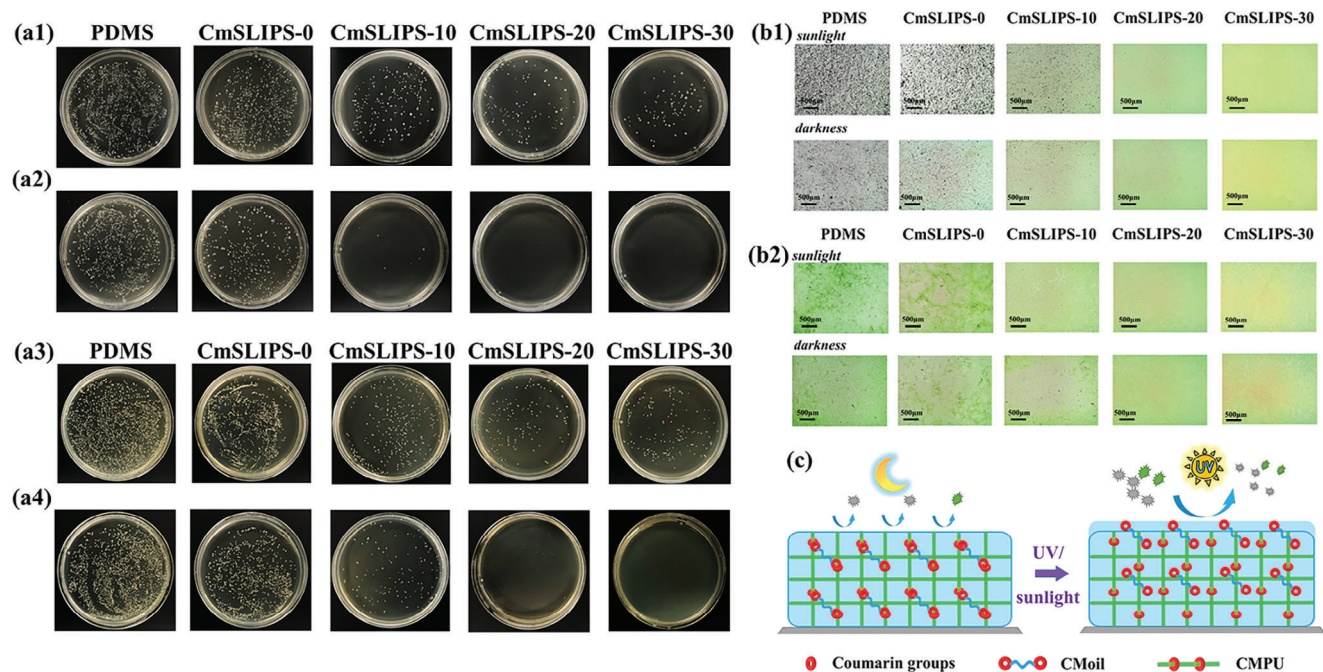


Figure 6. a1) Anti-bacteria performance: anti-*E. coli* test in the dark and a2) in the sunlight; a3) anti-*S. aureus* test in the dark and a4) in the sunlight; b1) anti-algae performance of different modes under different response conditions: "offensive" mode (diatom, sunlight) and "defensive" mode (diatom, darkness); b2) "offensive" mode (chlorococci, sunlight) and "defensive" mode (chlorococci, darkness); c) 2D diagram of antifouling mode responsively switching performance.

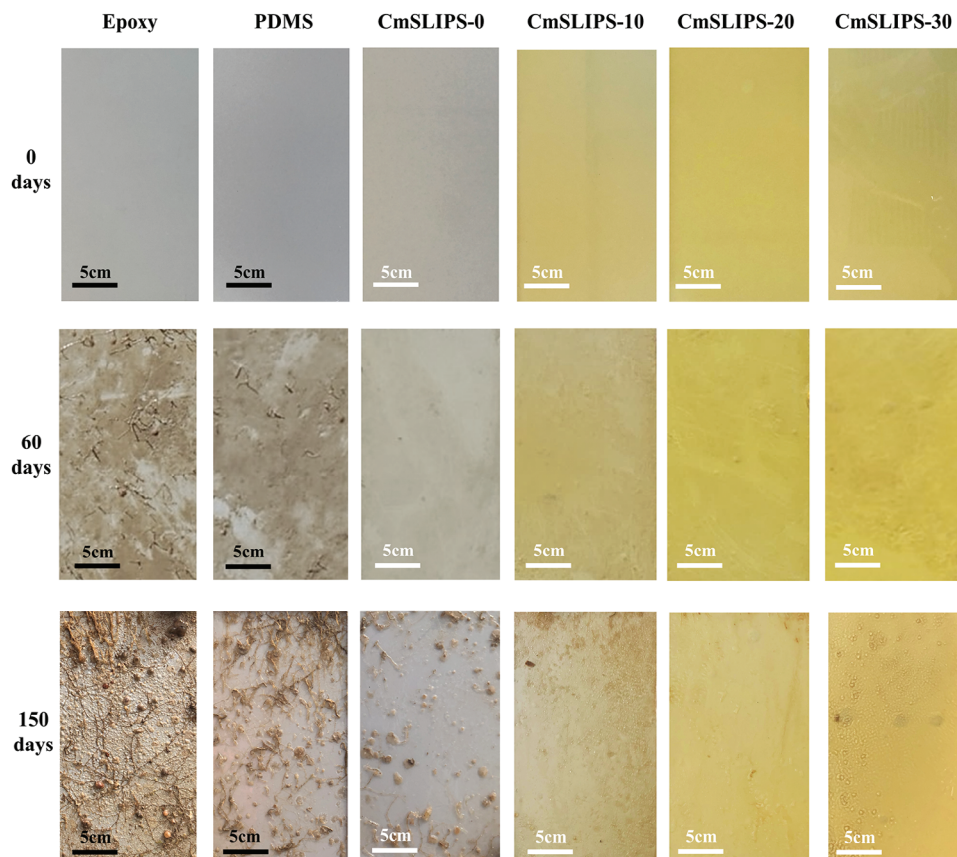


Figure 7. Marine field test of epoxy, PDMS and CmSLIPS-0/10/20/30 in the Zhoushan Sea for 150 days (2023.03-2023.08).

antibacterial effect of coumarin effectively enhanced the antifouling efficiency and intelligence of CmSLIPS.

The coatings were characterized by anti-algae and antibacterial tests under controlled culture conditions (Figure 6a,b). The PDMS and CmSLIPS-0 were unresponsive. The surface lubricity of CmSLIPS-10/20/30 after sunlight response was enhanced, and its antibacterial and anti-algae properties were enhanced. Under sunlight conditions, algae grow vigorously. The enhanced surface lubricating layer met the high antifouling demand at this time. Algae growth slowed down in the dark, while CmSLIPS without response was efficient enough for this situation. Although the model bacteria are not affected by light. But it still showed that the antifouling performance under light was better. As the CMPU and CMoil contained coumarin groups, they also had the antibacterial effect in the dark.

2.6.5. Pseudobarnacle Removal Performance and Marine Field Test

Barnacle is one of the common fouling organisms in the third stage of the marine biofouling process. Barnacles strongly adhere to solid surfaces and are difficult to be eradicated. Aluminum pseudobarnacles were used to imitate real barnacles to test the fouling release ability^[59] (Figure 5b). The removal shear strength of CmSLIPS was 0.20 Mpa, which was much lower than that of epoxy coatings (6.04 Mpa). Due to the low SE, low modulus and

dynamic lubricating layer, the pseudobarnacles did not attach to CmSLIPS firmly.

The antifouling property of CmSLIPS was tested in the East China Sea near the Zhoushan Islands for 150 d (2023.03-2023.08) in boom seasons. As shown in **Figure 7**, the epoxy, PDMS and CmSLIPS-0 coatings were adhered by amount of biofouling. There was no damage or biofouling on the CmSLIPS-10/20/30, which demonstrated the excellent real-sea antifouling performance in the long term. The antifouling effect was also strengthened as the content of CMoil increasing. The CmSLIPS exhibited excellent antifouling performance against inorganic pollutants, proteins, bacteria and algae, which effectively avoided the formation of biofilm in the former stages of the marine fouling process. Thus, the marine field antifouling test proved the practicality and feasibility of the CmSLIPS in the real neritic sea.

3. Conclusion

Herein, the bionic CmSLIPS based on coumarin was prepared as marine antifouling. Due to the reversible photodimerization of coumarin groups, the CmSLIPS adjusted the surface lubricity via the reversible “lock” of CMoil and CMPU. The adaptive adjustment of surface lubricity controlled the antifouling performance on demand. The coumarin antimicrobial groups enhanced the active biocidal antifouling property of SLIPS. The photodimerization “locking” mechanism of the lubricant enhanced the stability and durability in harsh environments, reduced the lubricant

loss and prolonged the service life. The CmSLIPS maintained satisfactory antifouling performance in the real neritic sea during 150 days in boom seasons. This study developed a bionic “smart” SLIPS coating, which is expected to expand its intellectualization in marine antifouling, biological antifouling and other fields.

4. Experimental Section

Details of the experimental procedures are given in the supporting information.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

bionic material, coumarin, lubrication adjustability, marine antifouling, reversible photo-dimerization

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